higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MATHEMATICS N5

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This marking guideline consists of 10 pages.

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QUESTION 1

1.1.1 $\ln y = \lim\limits_{x\to 0} \dfrac{\tan x - \sin x}{3x^2} \quad : \dfrac{0}{0}$

$= \lim\limits_{x\to 0} \dfrac{\sec^2 x + \cos x}{6x}$ ✓ $\quad : \dfrac{0}{0}$

$= \lim\limits_{x\to 0} \dfrac{2\sec x \,.\, \sec x.\tan x - \sin x}{6}$ ✓

$= 0$ ✓

(3)

1.1.2 $\ln y = 0$

$y = 1$ ✓ (1)

1.2 $f(x) = \dfrac{e^x - 1}{\sin x}, \quad [0°; 270°]$

$x = 0°$ ✓ $\quad or \quad x = 180°$ ✓ (2)

[6]

QUESTION 2

2.1 $f(x) = \cos x$

$f(x+h) = \cos x \cos h - \sin x \sin h$ ✓

$f'(x) = \lim\limits_{h\to 0} \dfrac{f(x+h) - f(x)}{h}$

$= \lim\limits_{h\to 0} \dfrac{\cos x \cos h - \sin x \sin h - \cos x}{h}$ ✓

$= \lim\limits_{h\to 0} \left[\cos x.\dfrac{\cos h - 1}{h} - \sin x \,.\, \dfrac{\sin h}{h}\right]$

$= \cos x.\lim\limits_{h\to 0} \dfrac{\cos h - 1}{h} - \sin x \,.\, \lim\limits_{h\to 0} \dfrac{\sin h}{h}$ ✓

$= \cos x \,.\, 0 - \sin x.1$ ✓

$= -\sin x$ (4)

2.2 2.2.1 $y = \sin^6(\ln(4x^5))$

$$\frac{dy}{dx} = 6\sin^5(\ln(4x^5))\checkmark \times \cos(\ln(4x^5))\checkmark \times \frac{1}{4x^5} \times 20x^4\checkmark \quad (3)$$

2.2.2 $$y = \left(\frac{x^2+4}{x^3-3x+6}\right)^4 + \sqrt{2x-5}$$

$$\frac{dy}{dx} = 4\left(\frac{x^2+4}{x^3-3x+6}\right)^3\checkmark \times \frac{2x(x^3-3x+6)\checkmark - (3x^2-3)(x^2+4)\checkmark}{(x^3-3x+6)^2\checkmark} + \frac{1}{2\sqrt{2x-5}} \times 2\checkmark \quad (5)$$

2.3 $y = (\cos x)^{-2x}$

$\ln y = -2x \ln\cos x\checkmark$

$$\frac{1}{y}\frac{dy}{dx}\checkmark = -2\ln\cos x\checkmark - 2x \times \frac{1}{\cos x} \times (-\sin x)\checkmark$$

$$\frac{1}{y}\frac{dy}{dx} = -2\ln\cos x + 2x\tan x$$

$$\frac{dy}{dx} = y(2x\tan x - 2\ln\cos x)\checkmark$$

$$\frac{dy}{dx} = (\cos x)^{-2x}(2x\tan x - 2\ln\cos x) \quad (5)$$

2.4 $\tan 3y - y^3\cos x + 3xy^2 = -4$

$$\sec^2 3y\frac{dy}{dx}\checkmark - \left[y^3(-\sin x) + 3y^2\cos x\frac{dy}{dx}\right]\checkmark + 3y^2 + 6xy\frac{dy}{dx}\checkmark = 0$$

$$\sec^2 3y\frac{dy}{dx} + y^3\sin x - 3y^2\cos x\frac{dy}{dx} + 3y^2 + 6xy\frac{dy}{dx} = 0$$

$$[\sec^2 3y - 3y^2\cos x + 6xy]\frac{dy}{dx} = -y^3\sin x - 3y^2\checkmark$$

$$\frac{dy}{dx} = \frac{-y^3\sin x - 3y^2}{\sec^2 3y - 3y^2\cos x + 6xy\checkmark} \quad (5)$$

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QUESTION 3

3.1 3.1.1

$$f(x) = x^3 - 3x^2 - 3x + 6$$
$$f'(x) = 3x^2 - 6x - 3 = 0$$
$$x^2 - 2x - 1 = 0$$
$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-1)}}{2(1)}$$
$$x = -0{,}414 \quad or \quad x = 2{,}414$$
$$y = 6{,}657 \quad ✓ \quad y = -4{,}657$$

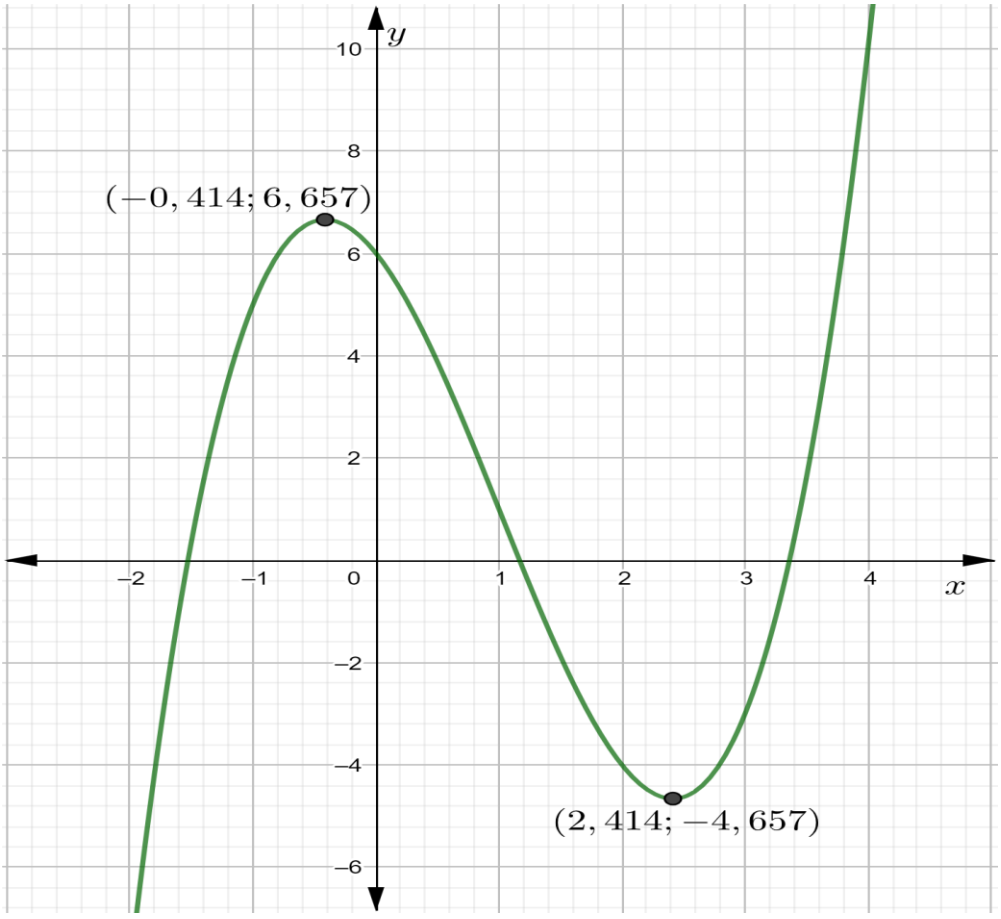
Thus, the coordinates of the turning points are $(-0{,}414\ ;\ 6{,}657)$ and $(2{,}414; -4{,}657)$.✓ (3)

3.1.2

x	-2	-1	0	1	2	3	4
y	-8	5	6	1	-4	-3	10

✓✓= 2 Marks for the any 2 correct answers (2)

3.1.3



✓ =1 mark for the shape
✓ =1 mark for the indication of the turning points on the graph (2)

3.1.4 Let $x_0 = -1,5$

$f(-1,5) = 0,375$

$f'(-1,5) = 12,750$✓

$x_1 = -1,5 - \frac{0,375}{12,750}$✓

$= -1,529$✓ (3)

3.2 3.2.1 $\frac{dV}{dt} = \frac{1}{12\pi}\left(2Ch\frac{dC}{dt}✓ + C^2\frac{dh}{dt}✓\right)$

This represents the rate of change of volume of the tree, or how fast the tree is growing.✓ (3)

3.2.2 $\frac{dV}{dt} = \frac{1}{12\pi}(2 \times 5 \times 22 \times 0,2 + 5^2 \times 4)$✓

$= \frac{1}{12\pi}(44 + 100)$✓

$= \frac{12}{\pi}\ or\ 3,820\ m/year$✓ (3)

3.3 $S = x + \frac{1}{x}$✓

$S' = 1 - \frac{1}{x^2} = 0$✓

$x^2 - 1 = 0$✓

$x = \pm 1$

$\therefore x = 1$✓ (4)
[20]

QUESTION 4

4.1 4.1.1 $\int \cos 3x \sin(-7x)\,dx$

$= \frac{1}{2}\int (\sin(-4x) - \sin 10x)\,dx$✓

$= \frac{1}{8}\cos 4x$✓ $+ \frac{1}{20}\cos 10x$✓ $+ C$ (3)

4.1.2 $\int (\sin^4 x \operatorname{cosec} x)\, dx$

$= \int \sin^3 dx$ ✓

$= \int (1 - \cos^2 x) \sin x\, dx$ ✓

$= \int \sin x\, dx - \int \cos^2 x \sin x\, dx$ ✓

$= -\cos x$✓ $+ \frac{\cos^2 x}{2}$✓ $+ C$ (5)

4.1.3 $\int \frac{x^3 - 3x^2 - 10x + 24}{x + 4} dx$

$= \int \left(x^2 - 7x + 18 - \frac{48}{x + 4}\right) dx$ ✓

$= \frac{1}{3}x^3$✓ $- \frac{7}{2}x^2$✓ $+ 18x$✓ $- 48\ln(x + 4)$✓ $+ C$ (5)

4.1.4 $\int x^3 \ln 4x\, dx$

$let\ u = \ln 4x$	$dv = x^3\, dx$
$du = \frac{1}{4x} \times 4dx$	$v = \frac{1}{4}x^4$ ✓
$= \frac{1}{4} dx$	

$\int x^3 \ln 4x\, dx = \frac{1}{4}x^4 . \ln 4x - \int \frac{1}{4}x^4 . \frac{1}{x} dx$✓

$= \frac{1}{4}x^4 . \ln 4x - \frac{1}{4}\int x^3\, dx$ ✓

$= \frac{1}{4}x^4 . \ln 4x - \frac{1}{16}x^4$✓ $+ C$ (4)

4.2 $$\int \frac{2x-1}{(x+1)^3}\,dx$$

$$\frac{2x-1}{(x+1)^2} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{(x+1)^3}$$

$$2x - 1 = A(x+1)^3 + B(x+1)^2 + \text{C}\checkmark$$

$$A = -1\checkmark$$
$$B = 3\checkmark$$
$$C = -3\checkmark$$

$$= \int \left[\frac{-1}{x+1} + \frac{3}{(x+1)^2} - \frac{3}{(x+1)^3}\right] dx$$

$$= -\ln(x+1)\checkmark - \frac{3}{(x+1)}\checkmark + \frac{3}{2(x+1)^2}\checkmark + C$$ (7)

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QUESTION 5

5.1 $$\int_0^2 f'(x)\,[f(x)]^2\,dx$$

$$let\ u = f(x)$$
$$du = f'(x) \quad \checkmark$$
$$u = f(0) = 1 \quad and \ \ u = f(2) = 5$$

$$= \int_1^5 u^2 du\ \checkmark$$

$$= \left[\frac{1}{3}u^3\right]_1^5\ \checkmark$$

$$= \frac{1}{3} \times 5^3 - \frac{1}{3} \times 1^3$$

$$= \frac{124}{3}\ \checkmark$$ (4)

5.2 $\int_0^{\infty} e^{-2t}.pdt$ ✓ $= p\int_0^{\infty} e^{-2t}dt$

$= p\left[-\frac{e^{-2t}}{2}\right]_0^{\infty}$ ✓

$= p\left(0+\frac{1}{2}\right)$ ✓

$= \frac{p}{2}$ (3)

5.3 5.3.1 $A = \int_a^b (y_1 - y_2)\,dx$

$= \int_{-1,577}^{-0,422} [-x-(3x^2+5x+2)]\,dx$ ✓

$= \int_{-1,577}^{-0,422} (-3x^2-6x-2)dx$

$= [-x^3-3x^2-2x]_{-1,577}^{0,422}$ ✓

$= 0,770units^2$ ✓ (3)

5.3.2 $V = \pi\int_a^b ({y_1}^2 - {y_2}^2)\,dx$

$= \pi\int_{-1,577}^{-0,422} [(-x)^2-(3x^2+5x+2)^2]\,dx$ ✓

$\pi\int_{-1,557}^{-0,422} (-9x^4-30x^3-26x^2-20x-4)dx$ ✓✓

$= \pi\left[-\frac{9}{5}x^5-\frac{15}{2}x^4-\frac{26}{3}x^3-10x^2-4x\right]_{-1,577}^{-0,422}$ ✓

$= 2,902units^3$ ✓ (5)

5.4 $m = \sigma A$

$dm = \sigma dA$

$dI_z = r^2$

$= x^2 \sigma dm$✓

$= x^2 \sigma 2\pi x\, dx$✓

$= 2\pi\sigma \int_0^{\frac{r}{2}} x^3\, dx$ ✓

$= 2\pi\sigma \left[\frac{1}{4}x^4\right]_0^{\frac{r}{2}}$ ✓

$= 2\pi\sigma \left[\frac{r^4}{64} - 0\right]$

$= \frac{1}{32}\pi\sigma r^4 units^4$✓ (5)

[20]

QUESTION 6

6.1 $\frac{dy}{dx} = \frac{2xy}{y^2 - 1} \quad y(2) = 1$

$(y^2 - 1)\, dy = 2xy\, dx$

$\left(y - \frac{1}{y}\right) dy = 2x\, dx$✓

$\frac{y^2}{2} - \ln y$✓ $= x^2 + C$ ✓

$\frac{1^2}{2} - \ln 1 = 2^2 + C$

$C = -\frac{7}{2}$ ✓

$\frac{y^2}{2} - \ln y = x^2 - \frac{7}{2}$✓ (5)

6.2

$$14 + 2x\frac{d^2y}{dx^2} = 2x^3 - 6x^2$$

$$\frac{d^2y}{dx^2} = x^2 - 3x - 7\checkmark$$

$$\frac{d\ y}{dx} = \frac{1}{3}x^3 - \frac{3}{2}x^2 - 7x + A\checkmark$$

$$y = \frac{1}{12}x^4 - \frac{1}{2}x^3 - \frac{7}{2}x^2 + Ax + B\checkmark$$

(3)
[8]

TOTAL: 100

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